event table

Parameter (Field)	Description	Value	Type	Default Logic
eventTypeCode	A coded representation of the type of a transportation event such as Departure, Arrival, or Proof of Delivery.	For example, in a road freight order, typically one of the following code values can occur: • Arrival: ARRIV_DEST • Delayed: DELAYED • Departure: DEPARTURE • Proof of Delivery: POD • Proof of Pickup: POPU • Loading Begin: LOAD_BEGIN • Loading End: LOAD_END • Coupling: COUPLING • Decoupling: DECOUPLING	String (20)	Follow the mapping of **Event Type** to **Event Code** in the **IDOC Integration** tab of the MM app.
eventReasonCode	A coded representation of the reason of a transportation event.	For example, in a road freight order, typically one of the following code values can occur: • Breakdown: BRKDWN • Others: OTHERS • Traffic: TRAFFC • Waiting at Location: WTGLOC	String (20)	Use the value of the current event's eventReasonCode field.
eventReasonText	A text string that contains the reason of a transportation event.	Free input	String (255)	Use the value of the current event's eventReasonText field.

Parameter (Field)	Description	Value	Type	Default Logic
`referenceEventTypeCode`	A coded representation of a specific milestone referring to another event during the transport execution. For example, a delayed event refers to an arrival event.	For example, in a road freight order, typically one of the following code values can occur: • Arrival: `ARRIV_DEST` • Departure: `DEPARTURE` • Proof of Delivery: `POD` • Proof of Pickup: `POPU` • Loading Begin: `LOAD_BEGIN` • Loading End: `LOAD_END` • Coupling: `COUPLING` • Decoupling: `DECOUPLING`	String (20)	For delayed events and events that inherit `GTTDelayedEvent`, follow the mapping of `refPlannedEventType` to **Event Code** in the **IDOC Integration** tab of the MM app. For other events, this field will be empty.
`recipientSystemLocationInternalID`	A proprietary identifier for a location where the even was reported.		String (20)	If the current event matches with a planned event, the value will be the planned event's `locationAltKey`. Otherwise, the value will be provided by the current event's `locationAltKey`.
`stopId`	A proprietary identifier for a stop where the event was reported.	The `stopID` in your TM in SAP S/4HANA system.	String (10)	The field will be empty.
`expectedDateTime`	The point in time when the event is expected to occur.	Use the function `toTimestamp` to get the value.	Timestamp	If the current event matches with a planned event, the value will be the planned event's `planBusinessTimestamp`. Otherwise, the value will be empty.
`estimatedDateTime`	The point in time when the event is estimated to occur.	Use the function `toTimestamp` to get the value.	Timestamp	The field will be empty.

Parameter (Field)	Description	Value	Type	Default Logic
`actualDateTime`	The point in time when the event occurs.	Use the function `toTimestamp` to get the value.	Timestamp	Use the value of the current event's `actualBusinessTimestamp` field.
`transportationOrderID`	A unique identifier for a transportation order.	The Freight Order Number in your TM in SAP S/4HANA system.	String (20)	Use the current event's tracking ID.
`forwardAttachments`	When sending a specific event to your TM in SAP S/4HANA, you can choose to send attachments with the event or not. **Note** If you use the event array as the input, this parameter will only take effect within its respective event object.	`true`: The event with all its attachments will be sent to TM in SAP S/4HANA. `false`: Only the event will be sent to TM in SAP S/4HANA.	N/A	If you use the event object as the input, the default value is true. If you use the event array as the input, the default value is false.
`contentText`	Corresponds to text in **Content** field of TM in SAP S/4HANA.	Free input	String (255)	Use the value of current event's `Note` fied.
`contentTextTypeCode`	A coded reprensentation of the type of the content text.	User defined in TM in SAP S/4HANA.	String (5)	The default value is "TOREM". If you want to use custom text types, see Maintain Custom Content Text Types.
`contentTextCreationDateTime`	The point in time when the content text is provided.	Use the function `toTimestamp` to get the value, such as 2020-11-03T05:47:25.402Z.	Timestamp	Use the value of the current event's `actualBusinessTimestamp` field.
`contentTextLanguageCode`	A coded representation of the language of the content text.	User defined in TM in SAP S/4HANA. For example: • "DE": German	String (9)	The default value is "EN".

Parameter (Field)	Description	Value	Type	Default Logic
		• "EN": English • "ES": Spanish • "FR": French • "JA": Japanese • "PT": Portuguese • "ZH": Chinese • "ZF": Chinese trad.		
`emissionValue`	A number indicating the amount of carbon emissions.	Free input	Decimal (31, 34)	
`emissionValueUom`	A string representing the unit of measure of carbon emissions.	Free input	String (3)	
`emissionNamespace`	An address representing the namespace of the emission node within a SOAP message.	Free input	String (255)	

Examples:

Forward a single actual event:

```
if ( "com.tenantname.gtt.app.sofvpauto.Shipment.Arrival" == actualEvent.eventType )
  {
      var destinationName = "TO_BF9_960";
      var msgObj = toObject("{}");
      msgObj.creationDateTime = toTimestamp("2020-11-03T05:47:25.402Z");
      msgObj.recipientBusinessSystemID= "BF9_960";
      msgObj.recipientPartyLBNNo = "10000786";
      msgObj.senderPartyInternalID = "LBNG4ZC02";
      msgObj.senderPartyStandardID = " 10000088";
      var eventObj = toObject("{}");
      eventObj.eventReasonText = "test event";
      eventObj.eventTypeCode = "ARRIV_DEST";//If the data of carbon emission need to be sent to TM,
      eventObj.recipientSystemLocationInternalID = "LBN-E2E-03";
      eventObj.contentText = "test note";
      eventObj.contentTextCreationDateTime = toTimestamp("2020-11-03T05:50:25.402Z");
      eventObj.contentTextLanguageCode = "EN";
```

```
        eventObj.contentTextTypeCode = "TOREM";
        eventObj.emissionValue = "1256";//Either all of emissionValue, emissionValueUom, and emissior
        eventObj.emissionValueUom = "KG";
        eventObj.emissionNamespace = "http://sap.com/xi/TMS/Global";
        forwardEventToTM(msgObj, eventObj, destinationName);
    }
```

Forward multiple actual events:

```
// forwarding estimated timestamp to TM starts
eventForwardToTM = true;
function getPlannedBusinessTime(etaEventType, etaLocationAltKey){
    var i_plannedEvents = 0;
    var z_plannedBusinessTimestamp;
    while (i_plannedEvents < trackedProcess.plannedEvents.size()) {
        if (trackedProcess.plannedEvents[i_plannedEvents].eventType == etaEventType &&
            trackedProcess.plannedEvents[i_plannedEvents].locationAltKey == etaLocationAltKey)
        {
            z_plannedBusinessTimestamp = trackedProcess.plannedEvents[i_plannedEvents].plannedBusir
        }
        i_plannedEvents = i_plannedEvents + 1;
    }
    return z_plannedBusinessTimestamp;
}
function generateEventObjects(){
    var etaEventObject = toObject("{}");
    etaEventObject.eventReasonText = "sync back estimated date time from GTT";
    innerEventTypeInfo = getEventTypeInfo(actualEvent.estimatedTimestamps[i].eventType, "en");
    if (innerEventTypeInfo != null && innerEventTypeInfo.idocMapping != null && innerEventTypeInfo.
        etaEventObject.eventTypeCode = "EVENT_EST_UPD";//If the data of carbon emission need to be
        innerLocationAltkey = parseLocationAltKey(actualEvent.estimatedTimestamps[i].locationAltKey
        etaEventObject.recipientSystemLocationInternalID = innerLocationAltkey.id;
        etaEventObject.estimatedDateTime = actualEvent.estimatedTimestamps[i].estimatedTimestamp;
        etaEventObject.referenceEventTypeCode = innerEventTypeInfo.idocMapping.erpEventCode;
        etaEventObject.expectedDateTime = getPlannedBusinessTime(actualEvent.estimatedTimestamps[i]
        etaEventObject.contentTextLanguageCode = "EN" ;
        etaEventObject.contentText = "note from e2a";
        etaEventObject.contentTextTypeCode = "TOREM"
        etaEventObject.contentTextCreationDateTime = toTimestamp("2024-09-12T08:18:34.280Z");
        etaEventObject.emissionValue = "1256";//Either all of emissionValue, emissionValueUom, and
        etaEventObject.emissionValueUom = "KG";
        etaEventObject.emissionNamespace = "http://sap.com/xi/TMS/Global";
        etaEventObject.forwardAttachments = true;//If the attachments of this event need to be sent
        eventObjects.push(etaEventObject);
    }
}
if (DELETION_EVENT != actualEvent.eventType && SHIPMENT_TP == trackedProcess.trackedProcessType
&& SHIPMENT_TRACKING_ID_TYPE == actualEvent.trackingIdType && TOR_SHIPMENT_TYPE == trackedProcess.s
    if (actualEvent.messageSourceType == "GTT:WS" || actualEvent.messageSourceType == "GTT:VP") {
        if (trackedProcess != null && eventForwardToTM && actualEvent.containsProperty("estimatedTi
            var i = 0;
            var innerEventTypeInfo;
            var innerLocationAltkey;
            var eventObjects = [];
```

```
            while (i < actualEvent.estimatedTimestamps.size()) {
                generateEventObjects();
                i = i + 1;
            }
            forwardEventToTM(msgObj, eventObjects, destinationTMName);
        }
    }
}
//forwarding estimated timestamp to TM ends
```

For more on how to forward events back to TM in SAP S/4HANA, see Synchronize Actual Events Back to TM in SAP S/4HANA.

updateInboundDeliveryInERP

Function **updateInboundDeliveryInERP** calls ERP SOAP interfaces to update the inbound delivery in SAP ERP. Currently, the only available field for updates is the Planned Delivery Date. For more, see Synchronize ETA of the Inbound Delivery Back to SAP S/4HANA.

! Restriction

If your SAP ERP system is already integrated with SAP Extended Warehouse Management (SAP EWM), this function will not operate.

Before using this function, you should first go to your subaccount on SAP BTP cockpit and configure a destination. In the navigation area, go to **Connectivity** > **Destinations** and select **Create Destination**. When setting up your destination, you can define the key-value pair in the **Additional Properties** section:

- Enter "**sap.lbn.gtt.header.<key>**" : "**<value>**". The key-value pair will be added to the header of the forward request.
- If your destination is protected by a CSRF token, add the following properties:

Property	Description
sap.lbn.gtt.csrf.header	Mandatory. The header is used to store the value of the CSRF token. For the header, you can provide your own one based on your server's configuration. In most cases, it should be "x-csrf-token".
sap.lbn.gtt.csrf.method	For the method, you can use `GET` or `HEAD`. The default is `GET`.
sap.lbn.gtt.csrf.url	The URL is used to fetch the CSRF token. The default is the main URL that you configured in the destination.

```
/**
* Update the inbound delivery in the ERP system.
*
* @optional param messageHeader (creationDateTime, recipientBusinessSystemID, recipientPartyLBNNo,
* @optional param event (trackingId, deliveryVersion, plannedDeliveryTime) - Used for building ever
* @param destinationUpdateInboundDeliveryInERP - The destination name.
*/
updateInboundDeliveryInERP (messageHeader, event, destinationUpdateInboundDeliveryInERP);
```

`messageHeader` table

Parameter (Field)	Description	Value	Default Logic

Parameter (Field)	Description	Value	Default Logic
creationDateTime	The point in time when the event is created.	Use the function toTimestamp to get the value, such as 2020-11-03T05:47:25.402Z.	Use the current UTC time.
recipientBusinessSystemID			
recipientPartyStandardID			
senderPartyInternalID			
senderPartyStandardID			

event table

Parameter (Field)	Description	Value
trackingId	An ID representation of the delivery number in the ERP system.	An existing inbound delivery number in the ERP system. Refer to the field VBELN in the table LIKP in the ERP system.
deliveryVersion	A number representation of the current version of the inbound delivery.	The delivery version formation for the specified inbound delivery. Refer to field DLV_VERSION in the table LIKP in the ERP system.
plannedDeliveryTime	The new planned delivery date.	Use the function toTimestamp to get the value.

For example:

```
if ( "com.tenantname.gtt.app.sofvpauto.InboundDelivery.InboundDeliveryEvent" == actualEvent.eventTy
{
    var destinationName = " TO_QW9_170_UPDATE_INBDLV";
    var msgObj = toObject("{}");
    msgObj.creationDateTime = "2020-11-03T05:47:25.402Z";
    msgObj.recipientBusinessSystemID= "QW9_170";
    msgObj.recipientPartyLBNNo = "10000786";
    msgObj.senderPartyInternalID = "LBNG4ZC02";
    msgObj.senderPartyStandardID = " 10000088";
    var eventObj = toObject("{}");
    eventObj.trackingId = "187019198";
    eventObj.deliveryVersion = "0001";
    eventObj.plannedDeliveryTime = "2021-05-15T08:00:00.213Z";
    updateInboundDeliveryInERP(msgObj, eventObj, destinationName);

}
```

Object Structure References

Example of trackedProcess Object

```
{
    "isCanceled": false,
    "freightOrderId": "test_61",
"cargoWeightUOM": "KG",
    "processStatus_code": "EARLY",
    "references": [],
    "senderPartyId": null,
    "plannedArrivalDateTime": "2020-01-11T08:00:00.213Z",
    "departureLocationId": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:Location:shippingPoint:2010",
    "shipperLbnId": "10000034",
    "id": "58978ad7-20b4-11ea-a0f2-51352b993cd3",
    "trafficDirection": "NOR",
    "numberOfUnloadingStops": 2,
    "eventReasonText": null,
    "partyId": "LBN#10000034",
    "integratedWithVP": null,
    "trackingId": "test_61",
    "longitude": null,
    "creationDateTime": "2019-12-17T10:02:34.075Z",
    "goodsValueCurr": "EUR",
    "messageSourceType": null,
    "actualBusinessTimeZone": null,
    "cargoQuantity": 5.00000000000000,
    "eventType": "com.sap.gtt.app.tfo.FreightOrderModel.FreightOrderEvent",
    "plannedArrivalDateTimeZone": "CET",
    "cloneInstanceId": "a6eeb2a8-3f82-456b-8863-5149572e604a",
    "cargoWeight": 70.00000000000000,
    "incoterm": "EXW",
    "nextStopId": null,
    "orderingPartyUserId": "10000034",
    "stops": [
        {
            "locationTimeZone": "UTC+3",
            "freightOrderEvent_id": "58978ad7-20b4-11ea-a0f2-51352b993cd3",
            "plannedDepartureAt": "2020-01-06T08:00:00.213Z",
            "plannedArrivalAt": "2020-01-06T08:00:00.213Z",
            "departureDateTimeDSTIndicator": null,
            "arrivalDateTimeDSTIndicator": null,
            "stopId": 30,
            "sequentialNumber": 3,
            "locationAltKey": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:Location:shippingPoint:201
        }
    ],
    "locationAltKey": null,
    "plannedDepartureDateTimeZone": "CET",
    "meansOfTransport": "Truck",
    "carrierExternalId": "LBN_CAR001",
    "departureStopId": 1,
    "scheme": "xri://sap.com/id",
    "logicalSystem": "Q8JCLNT774",
    "carrierLbnId": "10000012",
    "cargoVolumeUOM": "m3",
    "createdByUser": "gtt_v2_dev@sap.com",
    "latitude": null,
```

```
    "shippingType": "FTL",
    "goodsValue": 90.000000,
    "numberOfLoadingStops": 2,
    "cargoVolume": 12.00000000000000,
    "plannedTotalDistance": 200.000000,
    "registrationCountry": "DE",
    "vehicle": "0023133",
    "plannedDepartureDate": "2019-12-30",
    "plannedTotalDistanceUOM": null,
    "plannedNetDuration": "PT1H",
    "eventReasonCode": null,
    "altKey": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:FreightOrder:test_61",
    "originalDocuments": [
        {
            "number": "1234",
            "freightOrderEvent_id": "58978ad7-20b4-11ea-a0f2-51352b993cd3"
        }
    ],
    "plannedArrivalDate": "2019-12-06",
    "estimatedArrivalTime": "2020-01-06T08:00:00.213Z",
    "shipperExternalId": "LBN_SHIPP1",
    "plannedGrossDuration": "PT1H",
    "numberOfVisits": 5,
    "businessReferenceNo": [
        {
            "itemId": "item123",
            "freightOrderEvent_id": "58978ad7-20b4-11ea-a0f2-51352b993cd3",
            "documentTypeCode": "pdf",
            "itemTypeCode": "txt",
            "documentId": "refer1234"
        }
    ],
    "subaccountId": "86cbb49a-0deb-4c74-9c85-9a85908569f1",
    "actualTechnicalTimestamp": "2019-12-17T10:02:34.075Z",
    "dangerousGoods": "1",
    "plannedEvents": [
        {
            "isFinalPlannedEvent": false,
            "plannedBizTsEarliest": "2020-01-05T07:50:00.213Z",
            "plannedBusinessTimestamp": "2020-01-05T08:00:00.213Z",
            "latitude": null,
            "plannedBusinessTimeZone": "CET",
            "plannedTechnicalTimestamp": "2020-01-05T11:00:00.213Z",
            "eventType": "com.sap.gtt.app.tfo.FreightOrderModel.ArrivalEvent",
            "event_id": "58978ad7-20b4-11ea-a0f2-51352b993cd3",
            "id": "58978ad8-20b4-11ea-a0f2-85331debb9ae",
            "locationAltKey": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:Location:shippingPoint:201
            "plannedTechTsLatest": "2020-01-05T11:10:00.213Z",
            "plannedTechTsEarliest": "2020-01-05T10:50:00.213Z",
            "plannedBizTsLatest": "2020-01-05T08:10:00.213Z",
            "eventMatchKey": "30",
            "longitude": null
        }
    ],
```

```
    "cargoQuantityUOM": "PAL",
    "arrivalLocationId": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:Location:shippingPoint:2001",
    "plannedDepartureDateTime": "2019-12-30T08:00:00.213Z",
    "actualBusinessTimestamp": "2019-12-29T08:00:00.213Z",
    "estimatedArrivalTimeZone": null,
    "registrationNumber": "HDG123",
    "trackingIdType": "FreightOrder",
    "arrivalStopId": 40,
    "eventMatchKey": null
}
```


Example of actualEvent Object


```
{
    "scheme": "xri://sap.com/id",
    "logicalSystem": "Q8JCLNT774",
    "references": [],
    "createdByUser": "gtt_v2_dev@sap.com",
    "latitude": 50.123600000,
    "senderPartyId": null,
    "id": "bf1a5455-20b3-11ea-a0f2-c95cd3b8a2f4",
    "eventReasonCode": null,
    "altKey": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:FreightOrder:test_676",
    "eventReasonText": "test_676ArrivalEvent_INTERMEDIATE",
    "partyId": "LBN#10000034",
    "estimatedArrivalTime": null,
    "trackingId": "test_676",
    "longitude": 100.123000000,
    "creationDateTime": "2019-12-17T09:58:16.563Z",
    "subaccountId": "86cbb49a-0deb-4c74-9c85-9a85908569f1",
    "messageSourceType": null,
    "actualTechnicalTimestamp": "2019-12-17T09:58:16.563Z",
    "actualBusinessTimeZone": "UTC+8",
    "eventType": "com.sap.gtt.app.tfo.FreightOrderModel.ArrivalEvent",
    "cloneInstanceId": "a6eeb2a8-3f82-456b-8863-5149572e604a",
    "actualBusinessTimestamp": "2019-12-12T10:00:00.213Z",
    "estimatedArrivalTimeZone": null,
    "trackingIdType": "FreightOrder",
    "nextStopId": null,
    "locationAltKey": "xri://sap.com/id:LBN#10000034:Q8JCLNT774:Location:shippingPoint:2011",
    "eventMatchKey": "30"
}
```


Example of plannedEvent Object


```
{
    "isFinalPlannedEvent": false,
    "lastProcessEventDirectory_id": "bf335a96-20b3-11ea-a0f2-4133d7ffdbfd",
    "process_id": "f0981857-f825-5a24-a664-f3b57fda7ae5",
    "nextOverdueDetection": "2020-01-05T11:10:00.213Z",
```